# Paper-Based Colorimetric Detection of Iron Contamination in Drinking Water

# Iron Contamination: A Global Challenge

Iron contamination in drinking water is a widespread issue, especially in regions dependent on groundwater and old pipeline systems. Corrosion of iron pipes and natural iron-rich aquifers lead to elevated iron levels in water.

Although iron is an essential nutrient, excess iron in drinking water causes:

- Metallic taste and unpleasant odor
- Staining of clothes and utensils
- Long-term health concerns when consumed continuously

Testing iron in water generally requires laboratory-based instruments, which are costly and inaccessible in rural and resource-limited areas.

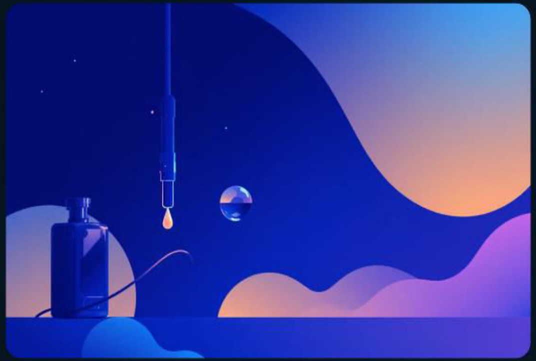
# Why This Problem Is Important

## Water Quality & Public Health

Iron contamination directly affects water quality and public health.

## Unknowing Consumption

Many households consume iron-contaminated water unknowingly.



## Need for Screening Tools

There is a strong need for rapid, low-cost, on-site screening tools.

## Early Detection Benefits

Early detection enables timely filtration and corrective measures.

# Our Biotechnology Approach

As biotechnology students, we are trained to apply chemical and biological principles to solve real-world problems. This project aligns with our field because:

- Metal ion detection is a key application of analytical biotechnology
- Colorimetric sensing forms the basis of many biosensors
- Biotechnology enables translation of lab chemistry into portable diagnostic tools

Our goal was to design a simple yet effective detection system suitable for community-level use.

# Prototype Description: Paper-Based Colorimetric Iron Detection Kit

## What We Developed

A paper-based sensor treated with an iron-reactive reagent that produces a visible color change when dipped in iron-containing water.

## Key Features

- No electricity required
- Rapid detection (within seconds)
- Easy visual interpretation
- Low cost and portable
- Suitable for field testing

# Working Principle: Colorimetric Reaction

The detection is based on the formation of a colored coordination complex between ferric ions and thiocyanate ions.

$$Fe^{3+} + SCN^{-} \rightarrow \text{Ferric thiocyanate (reddish-orange complex)}$$

As iron concentration increases:

- More complex is formed
- Color intensity increases

This allows visual estimation of iron levels.

# Experimental Design: Simulated Water Samples

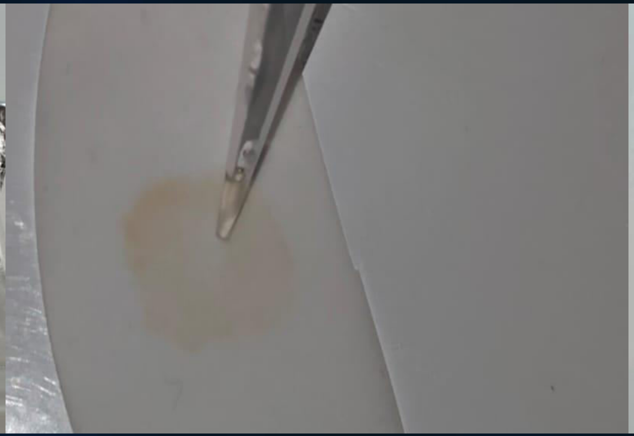
To demonstrate the detection principle, we prepared simulated drinking water samples:

| Sample | Iron Concentration | Interpretation |
| --- | --- | --- |
| Sample A | 5 mg/L | Low iron |
| Sample B | 10 mg/L | Moderate iron |
| Sample C | 15 mg/L | High iron |

These concentrations were optimized to ensure clear naked-eye visibility.

## Sample Preparation

Prepare the iron water samples in separate test tubes along with the reagent and stock



## Detection Method

Dip thiocyanate-treated strip into each sample and check the colour change

## Iron Observation

Observe color changes: faint orange (low iron) to deep redish oranage (high iron).

## Filter paper observation

Control strip in distilled water shows no color change whereas the others show a colour gradient

# Why We Used Optimized Concentrations (5 : 10 : 15)

Actual drinking water iron levels are very low. For demonstration and visual detection, concentrations were amplified.

Optimization ensures:

- Clear color contrast
- Avoidance of faint or saturated signals

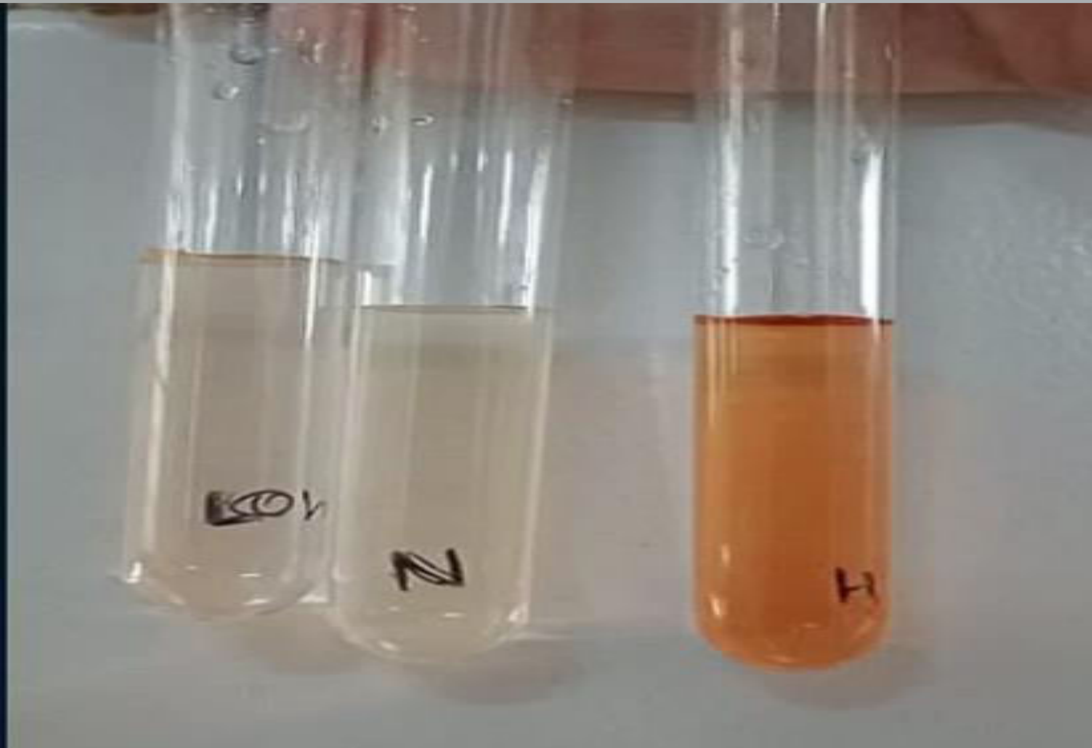
This process is known as assay optimization.

## Limitations of the Current Prototype

- Semi-quantitative (visual estimation only)
- Uses simulated samples
- Sensitivity not yet at regulatory limits

# Results & Observations

- A clear color gradient was observed across the samples.
- Color intensity increased proportionally with iron concentration.
- The test provided rapid and reproducible visual results.
- Demonstrates feasibility of paper-based iron detection.

# Why Monitoring Iron in Wastewater Is Important

### 1 Industrial Release

Industrial plants release wastewater containing dissolved iron from processes, pipes, and corrosion.

### 2 Health Effects of Excess Iron

Excess iron causes reddish-brown water, bad odor and taste, and soil and surface water contamination.

### 3 Common Uses Near Industries

People living near industries often use this water for washing clothes and utensils, cooking and domestic activities, and irrigation of nearby land.

### 4 Long-term Exposure Risks

Long-term exposure may lead to skin and hair problems, reduced water usability, and environmental damage.

Early detection of iron helps prevent long-term health and environmental effects.

# Future Application – Wastewater Screening Using 5:10:15 Ratio

Industrial wastewater often contains higher iron concentrations than drinking water. Our 5:10:15 ppm iron experiment represents a practical range for wastewater monitoring.

| | |
|---|---|
| **5 ppm**<br>Moderate contamination | **10 ppm**<br>High contamination |
| **15 ppm**<br>Severe contamination | |

## This Range Enables:

- Clear visual detection without instruments
- Rapid on-site screening near industrial zones
- Awareness of contamination severity

## Communities and Authorities Can:

- Decide when treatment is required
- Compare iron levels before and after treatment

Thus, our kit can evolve into a simple wastewater monitoring and awareness tool.

# Future Scope & Advancements

### Enhanced Sensitivity

Optimize reagent loading to detect lower iron levels.

### Quantitative Analysis

Integrate smartphone-based color analysis for precise measurements.

### Kit Miniaturization

Develop sealed, disposable paper strips for portability.

### Real Sample Testing

Apply kit to household and groundwater for accuracy.

### Community Deployment

Deploy in rural areas for screening and awareness.

# SDG Relevance




## SDG 3 – Good Health and Well-Being

Ensures safe drinking water

Reduces long-term health risks

## SDG 6 – Clean Water and Sanitation

Promotes water quality monitoring

## SDG 10 – Reduced Inequalities

Affordable diagnostics for underserved communities

# Final Conclusion

Our prototype demonstrates how a simple chemical reaction can be transformed into a low-cost, portable water testing tool.

This project validates the signal-generation principle of iron detection and provides a strong foundation for future development of accessible water quality diagnostics. This project demonstrates a rapid, paper-based approach for iron detection in water, offering a scalable solution for community-level water quality screening.

# THANK YOU

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